

DOE SBIR/STTR FY26 PHASE I • GENESIS MISSION

Nothing to sign at the pitch stage

DOE requires no forms, certifications, or representations at the pitch stage. The only gate is an AMP account. Stage 2 applies only if the pitch is invited forward, and DOE has not yet published the full-application forms package. Every Stage 2 row is marked by source so nothing anticipated is mistaken for something confirmed.

CONFIRMED

Stated on a DOE or ConnectWerx page as of August 22, 2026

ANTICIPATED

Standard SBIR/STTR practice, not yet published here

STAGE 1 • PITCH**Due Sep 10, 2026, 2:00 PM ET****REQUIRED****AMP account request**

WHERE register.ati.org | SIGNS Michael Ward, no signature

Required before submission. Under 5 minutes to request, allow 1 to 2 business days for approval.

READY**Pitch submission form, four open-text answers**

WHERE AMP, sbirsttr-amp.ati.org | SIGNS No signature

Text is written. See the Pitch Responses document in this package.

REQUIRED**Topic selection, one only**

WHERE AMP submission form | SIGNS No signature

Topic 1, Scaling the Biotechnology Revolution. Cannot be changed after submission.

OPTIONAL**Bibliography and References Cited**

WHERE AMP file upload, PDF, 5 MB or smaller | SIGNS No signature

Prepared. Upload the file named PlastiBioFuelBibliography.pdf.

NOT APPLICABLE**Certifications and representations**

WHERE Not applicable | SIGNS Not applicable

None required at pitch stage. Confirmed against the ConnectWerx opportunity page and FAQ on August 22, 2026.

NOT APPLICABLE

SBIR versus STTR election

WHERE Not applicable | SIGNS Not applicable

Not made at pitch. DOE states pitches are not evaluated separately for SBIR and STTR.

STAGE 2 · FULL APPLICATION

Only if invited

CONFIRMED

Active SAM.gov registration and UEI

WHERE sam.gov/entity-registration | SIGNS Michael Ward as Entity Administrator
UEI KQAWZ54RDUM8 must be entered on the application.

CONFIRMED

SBA Company Registry registration and proof

WHERE app.www.sbir.gov/company-registration | SIGNS Michael Ward
Proof of registration is required as part of the application.

CONFIRMED

Small Business Concern eligibility representation

WHERE Full application | SIGNS Michael Ward, Founder and CEO
For-profit, U.S. place of business, 51 percent or more U.S. citizen owned and controlled, 500 or fewer employees. Form not yet published.

CONFIRMED

Place of performance

WHERE Full application | SIGNS Michael Ward
All research and development performed in the United States.

CONFIRMED

SBIR only: PI primary employment certification

WHERE Full application | SIGNS Michael Ward as PI
More than 50 percent employment with the small business.

CONFIRMED

SBIR only: work-share certification

WHERE Full application | SIGNS Michael Ward
Small business performs at least two-thirds of Phase I effort.

CONFIRMED

STTR only: work-share certification

WHERE Full application | SIGNS Michael Ward and UNT authorized official
Small business at least 40 percent and research institution at least 30 percent.

CONFIRMED

STTR only: written IP allocation and licensing agreement

WHERE Executed before work begins | SIGNS Michael Ward and UNT authorized official
DOE states this must be in place before work begins.

ANTICIPATED

Essentially equivalent work and duplicate funding disclosure

WHERE Full application | SIGNS Michael Ward
Standard across SBIR and STTR agencies. Ready-to-use language is below.

ANTICIPATED

Current and pending support for all personnel

WHERE Full application | SIGNS Michael Ward
Not yet published for this opportunity.

ANTICIPATED**UNT subaward statement of work and budget**

WHERE Full application attachment | SIGNS Prepared by PlastiBioFuel with UNT input
Award-stage under 2 CFR 200.332. Institutional sign-off is not required at submission.

ANTICIPATED**Budget and line-by-line justification**

WHERE Full application | SIGNS Michael Ward
Must not exceed \$250,000, or \$256,500 with TABA.

ANTICIPATED**Foreign risk management and research security disclosures**

WHERE Full application | SIGNS Michael Ward
DOE maintains a Foreign Risk Management page for the SBIR and STTR programs.

DUPLICATE FUNDING DISCLOSURE, READY TO USE

Use this wording on any essentially equivalent work certification at full application. It is already reflected in Question 4 of the pitch.

“PlastiBioFuel LLC has one other federal proposal pending: NSF SBIR Phase I, Research.gov proposal 328095, solicitation NSF 26-510, subtopic BT10 Synthetic Biology and Metabolic Engineering, total requested \$304,955, submitting against the November 4, 2026 window. That proposal addresses COF-enzyme reusability, the monomer-to-ethanol biological conversion pathway, and construction of an integrated lab-scale spiral reactor module. The scope proposed to DOE is the computational design layer: a physics-informed machine learning surrogate model of the immobilized-enzyme reactor and its experimental validation. The two efforts share no tasks, no personnel effort months, and no budget lines. This is not essentially equivalent work, and no duplicate funding would result from award of both.”

GENERATIVE AI USE DISCLOSURE, READY TO USE

DOE asks every applicant to state the extent of generative AI use and how it was used. Paste this into that field. It is written for a reader who must not confuse the two uses: AI is the engineering method behind the technology, and it was separately used to draft this text.

Generative AI was used in two distinct ways, and they should be read separately.

1. In the research itself. Michael David Ward, Founder and CEO and the proposed principal investigator, used a commercial large language model (Grok, xAI) as a modeling and design environment to derive and test the continuous spiral-flow reactor this proposal describes. Channel geometry, flow behavior, and enzyme reusability were explored across more than one million simulation runs. He built that capability and operates it himself. This is the subject of the proposal rather than an aid to writing it, and the Phase I work proposed here is the work of quantifying where that model is trustworthy against physical measurement. The model has not been validated against a physical reactor, and the proposal says so.
2. In preparing the written submission. A commercial generative AI assistant (Claude, Anthropic) was used to draft, organize, and format the narrative text of this submission from source material supplied by the applicant: the company technical file, the University of North Texas Final Project Report for Phase-I signed by Prof. Shengqian Ma and dated August 7, 2026, correspondence from Dr. Bradley Phipps, USPTO provisional application 63/848,456, and the agency's own published solicitation pages.

What generative AI did not do. It did not generate, estimate, or extrapolate any experimental result. Every laboratory value here, including the 122 $\mu\text{mol/h/mg}$ immobilized turnover figure, comes from the signed UNT report. The one figure sourced outside that report, roughly 3.6 times the standard rate at 80 °C, is attributed to Dr. Phipps by email and is stated with its confounding caveat. Every reference was verified against PubMed by identifier before inclusion. No text, data, or citation was accepted without tracing it to a named primary source.

Responsibility. Michael David Ward reviewed and approved every statement in this submission and is responsible for its accuracy, authenticity, and authorship. PlastiBioFuel LLC is fully capable of performing the work described. To the best of the applicant's knowledge, this submission does not infringe or violate the rights of any third party.

TWO STANDING RULES FOR THIS PACKAGE

1. Dr. Shengqian Ma is Other Personnel on NSF-style forms, not Senior or Key Personnel. DOE has not published its personnel categories for this opportunity, so apply the same treatment unless the DOE form defines categories differently.
2. ARG Petro is offtake discussions and interest. There is no signed letter of intent and no executed offtake agreement. No document in this package may say otherwise.